

Automotive intelligence: Unleashing the potential of AI beyond advance driver assisting system, a comprehensive review

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ABSTRACT

Big data, AI, and machine learning have ushered in a transformative era in computer processing and application. AI's capacity to emulate human cognition spans a spectrum of industries, from rudimentary tasks to intricate decision-making processes. Projections indicate that within a short span, AI technology will bestow computers with human-like capabilities, rendering it indispensable. This advancement has substantially bolstered industry autonomy, fortifying security and safety through the integration of electronics, internet connectivity, advanced algorithms, and cutting-edge processing chips.

In this context, Automated Guided Vehicles (AGVs) leverage AI-powered systems and sensors to optimize performance and proactively thwart potential failures. Going beyond Advanced Driver Assistance Systems (ADAS), this review scrutinizes the broader spectrum of AI applications. Specifically, it synthesizes pertinent studies encompassing AI's impact on car emissions, predictive maintenance, security, and the realm of connected vehicles.

While AGVs can utilize AI algorithms to chart emission-efficient routes and engineer energy-conservative vehicles, AI, through diverse algorithms, proficiently anticipates component failures, substantially enhancing the cost-effectiveness of predictive maintenance and bolstering vehicle reliability. Moreover, AI integrated with sensor technology safeguards vehicles against a spectrum of threats. Neural networks and machine learning further automate security measures for both cargo and passengers. Concurrently, AI and machine learning fortify traffic control and enhance vehicle communication within connected vehicle frameworks, albeit necessitating fortifications in data security, encryption, and system efficiency.

This survey paper comprehensively covers the expanse of automotive AI and machine learning applications, extending beyond ADAS. It underscores that the evolving mobility landscape necessitates a thorough evaluation of network architecture, connectivity, and performance metrics, encompassing AGV technology, networking protocols, and swarm dynamics. In this vein, the paper advocates for experts to address existing literature gaps and proactively enhance automotive systems.

1. Introduction

The scope of computer processing and its practical application has significantly expanded due to the emergence of big data,

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artificial intelligence (AI), and machine learning (ML). AI is ubiquitously employed across various industries, encompassing both rudimentary task completion and intricate decision-making processes [1]. As technological advancements continue to propel the AI domain forward, the ultimate goal is to develop computers capable of emulating human cognition and behaviour.

Every new car produced is smarter, sensor-based, and powered by artificial intelligence. And frequent inspections of these sensors are required to ensure that the systems do not malfunction and interfere with the journey. We have developed several neural networks, machine learning methods, and algorithms to replace manual driving and this work is currently ongoing. To guarantee that these systems are precise and reliable, we have relied largely on using sensors and mechatronic components.

All this work on developing autonomous guided vehicles ensures their drivability and safety. Nevertheless, a vehicle is much more than that, and the major goal of this review article is to apply artificial intelligence to other automotive areas. This paper has combined many studies to better understand automotive research outside ADAS (Advance Driver Assisting System). Beginning with car emissions, artificial intelligence calculates emissions to forecast and attempt to set the system to limit emissions. The quantity of CO₂ discharge may be predicted given parameters like distance, fuel type, time, traffic, etc., allowing the autonomous guided vehicle to choose an effective way to emit fewer emissions and allowing corporations to design better, more energy-efficient vehicles from these data [2,3]. The health of components of automotive systems may also be predicted using artificial intelligence, which helps us determine when to do predictive maintenance and which parts need to be repaired [4]. Various algorithms, including fuzzy logic, artificial neural networks, GPR, k-NN, SVM, and others, are used to process the data, categorize the features, and produce the results of predictive maintenance [5].

To save costs and boost efficiency, the driver can have their car serviced, or the AGV can service the vehicles. The developed Artificial Intelligence model demonstrates the effectiveness of using artificial intelligence for the vehicle's security, making it more resistant to assaults. With the use of neural networks and machine learning, these security features will defend the cargo or the human from external threats when we fully rely on AGVs in the future. With respect to security and communication, using machine learning and internet implementation, a new field of linked vehicles is created, where communication between vehicles will improve the effectiveness of traffic control, among other things [6]. However, the increasing connectivity and intercommunication between vehicles in the context of linked automobiles raise significant safety concerns due to the potential vulnerability to hacking attempts and subsequent accidents caused by malicious hackers. Furthermore, the accessibility of the vehicle's data poses an additional security risk. Nevertheless, it is crucial to acknowledge that future advancements in the automotive industry will focus on enhancing the system's efficiency, implementing robust encryption measures, and bolstering overall security protocols. These advancements aim to ensure that sensitive data remains protected from unauthorized access or compromise [7,8].

Moreover, linked automobiles make autonomous driving possible, which means fewer sensors will be used. Even though today's AGVs have semi-autonomous control, we still require a driver to take charge in complicated scenarios when the artificial intelligence model cannot make an appropriate judgement. Regarding the driver's mental load and delay the AI model can forecast the driver's behaviour based on that load and take action to prevent any fatal consequences of the driver's irresponsibility. To evaluate the behaviour, many image-processing neural networks, as well as sensors like LiDAR and cameras, have been incorporated. The model effectively forecasts wise choices .

This survey paper aims to cover the application of AI and machine learning in the automotive industry outside of ADAS systems. With an emphasis on describing the effect of mobility on network architecture, connection, and networking performance metrics, continuously evaluating the aspects of AGV networks, including AGV technology, networking protocols, and swarms. The material about the protocols, algorithms, neural networks, machine learning models, and most recent developments in each application domain was assembled. Discovered the gaps in the literature for each and reported them in the paper. So that experts might try to close the gap and improve the systems in the future.

2. Vehicle maintenance

Recent developments in maintenance modelling have made it possible for a wide range of applications, thanks in part to data-driven techniques like Machine Learning (ML). The car industry has been more concerned with maintaining functional safety while lowering maintenance costs. To do this, Predictive Maintenance is a crucial strategy (PdM). ML is a great option for PdM since modern vehicle produce a lot of operational data. In ML-based PdM for automotive systems, the discovery or mining technique uses reinforcement learning to identify the actions that produce the highest rewards. Modelling the system's typical behaviour and spotting deviations, or anomalies, which can be signs of current or developing defects, are key components of predictive maintenance (PdM).

Different models were compared from logistic regression to SVM and ANN with major objective to construct an intelligible model in identifying oil samples from diesel engines using a real-world case study. The characteristics, which ranged from obvious pollution to healthy conditions, were collected from off-road vehicles employed in the mineral extraction industry and categorized. Unsupervised machine-learning algorithms implemented using fleet data to locate faulty cells in electric car batteries.

The most reliable unsupervised learning method is a clustering approach, which studies various data to find hidden patterns or groupings. Several clustering methods include HDBSCAN, multilevel cluster analysis, fuzzy c-means, DBSCAN, and K-means. Classification approaches employ data marked to discover an input to class label mapping to training some decision functions. Naive Bayes classifiers, support vector machines (SVMs), and k-nearest neighbours (k-NN). Some common predictors are deep neural networks, random forests, decision trees, and XGBoost. A type of supervised algorithm known as regression uses a combination of data entering and leaving to teach itself how to forecast perpetual outputs from freshly acquired data.

2.1. Faults in the engine and transmission

Novel strategy for detecting engine defects using a supervised classification setup, a group of extraordinary Bayesian machines of learning (ELM), etc. like engine failure detection using vibration signals. An unique ANN variation for this purpose is theorized. An integrated method which is based on the study of harmonics using discrete wavelet transforms, rapid Fourier transforms, and artificial neural networks (ANNs) is being used as a hybrid technique for detecting combustion faults in a diesel engine with 12 pistons. Using a supervised machine learning framework, gearbox data can be categorized into three categories: broken, 10 % deteriorated, and 5 % deteriorated. Statistical features are retrieved from the vibration signals after undergoing a continuous wavelet transform. The classification of common issues in planetary gearboxes is done using a unique architecture called an amalgamated Deep Belief Network (DBN), such as (partially) fractured teeth. The results demonstrated that hybrid DBN outperformed other techniques.

2.2. Suspension and brake system faults

To simulate the motorcycle's rear suspension's pulse sensor, a delicate sensor called "Nonlinear Auto-Regressive with Exogenous Inputs" (NARX) and an ANN are employed [9]. The CNN attained classification precision of up to 92.22 %, contrasted to the ANN's 87.27 % when the results were compared to those of a conventional ANN. An accelerometer-equipped brake test rig is used to capture shaking information. From these signals, statistical features are extracted, and an algorithmic subset is selected. The categorization process is then carried out using a clonal selection method. The trembling data from the brake test equipment was preprocessed using wavelet analysis then this information is used to evaluate a variety of classification techniques, such as best first trees, an ANN, SVMs, and Hoeffding trees, to identify and categorize the number of different hydraulic brake phases, from "excellent" to "Drum Brake Pad Wear," "Air in Brake Fluid," and failure states like "Brake Oil Spill" and "Reservoir Leak." With the help of Hoeffding trees, the best categorization outcomes were achieved.

2.3. Electric vehicle and induction motor faults

The crucial step is characteristic extraction and the choice among two health gauges utilizing correlation modelling from the battery's internal resistance. Different technique were proposed for assessing battery conditions in real-world driving patterns. Utilizing measurable data from electric vehicles, such as current (I) and voltage (V), their statistics-driven approach monitors the State of Health. They trained a supervised random forest classifier. The test result was good and acceptable. The study employed an Artificial

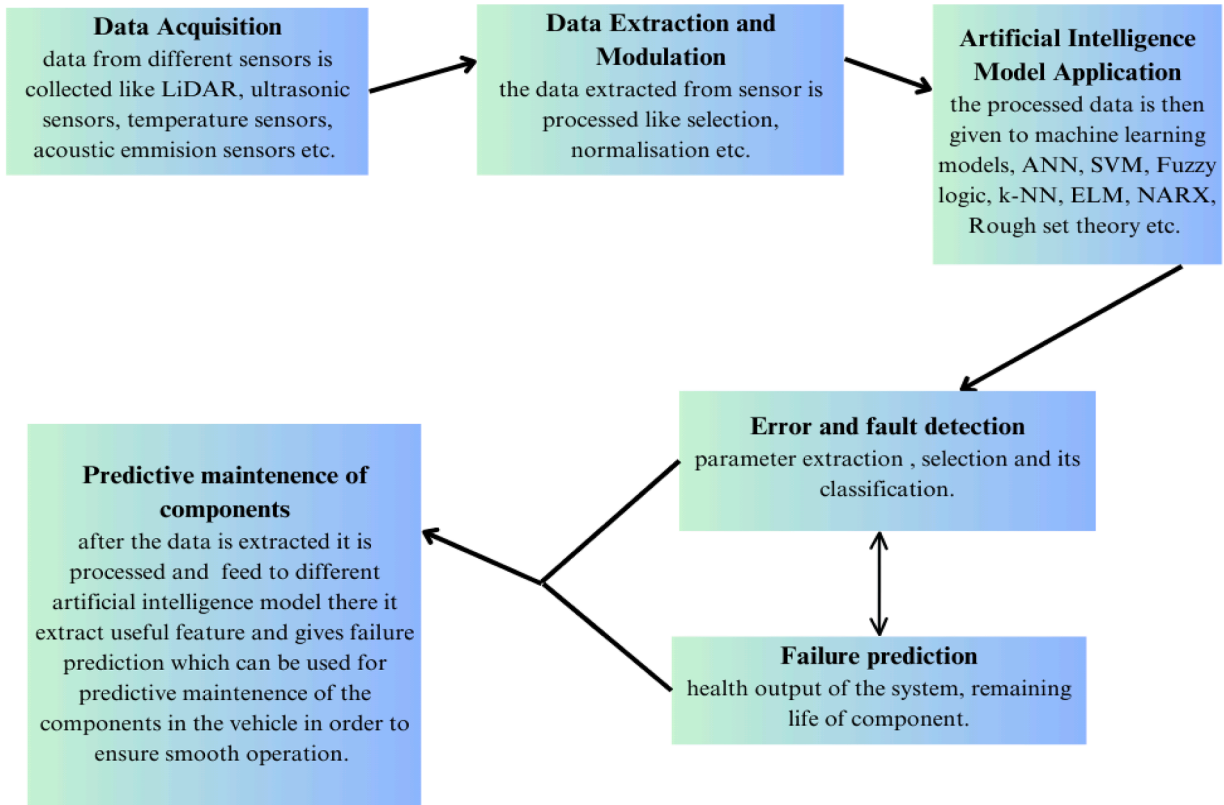


Fig. 1. Conceptual illustration of Predictive maintenance.

Neural Network (ANN) to develop a model with the three most critical variables, temperature and power factor. The temperature, vibration, and power factor models were validated with the motor operating at 38 °C to 41 °C, up to 3.8 Vibrations, and 0 to 100 kW, respectively. The significance of the investigation demonstrates that the MIF-based HVIM predictive maintenance model with ANN is suitable for industrial applications and that, in the future, industries or corporations may track and manage AGV fleets [10].

Nevertheless, the disadvantage of this research work is that the data utilized to train models are simulated or assumed data, the implication of real data is limited in number, and the validation of these datasets is limited. Another issue is that measuring the effectiveness of the proposed algorithms using real-world data is difficult. Real data is frequently only partially or not at all categorized, and annotating it requires specialized knowledge and skills. The potential Visual Interactive Labeling (VIAL) technique will be studied further in the future.

Fig. 1 presents the conceptual framework of predictive maintenance.

3. Vehicle security

With the advent of the internet, the reduction in the size of sensors and the availability of low-cost electronics gave rise to IoT; now, everything is interconnected to other intelligent devices via the internet, which creates a security issue making it easy for outsiders to attack. In addition, as the future relies significantly on internet-connected automobiles, ADAS system security must be updated to that level to assure safety. Some research has developed systems incorporating Artificial Intelligence, Machine Learning, and Neural Networks to encrypt sensitive data and make systems more durable and safer. In [11] proposed a simplified control competency and smooth integration, and the technique was exhibited on a Jetson AGX Xavier. It can attain up to 96 % accuracy. As a result, the proposed innovation is likely to increase safety and security in future independent open travel frameworks. Vehicle's reliance on vehicular ad hoc devices for outer communication is growing. The security component's Proportional Overlapping Scores technique enables the reduction of the number of highlights inside the Kyoto benchmark dataset. This cross-breed discovering framework employing back-engendering neural systems recognizes Dissent of Benefit (DoS), a prevalent attack method in automobile ad hoc networks. The result showed that the recommended interruption detection framework might be able to distinguish between hazardous and non-harmful vehicles in fully autonomous and even self-driving cars. Fig. 2 illustrates the functioning of an Artificial Neural Network (ANN) trained with the Kyoto Dataset, showcasing its mechanism in processing the data.

In Multi-Layer Perceptron (MLP), a group of non-linear neurons is linked to form a feed-forward multi-layer, the usefulness of this strategy in solving prediction and classification issues is widely established. Based on the experimental design of the IDS, crossover validation is used to determine the 'optimal' quantity of layers that are concealed and neurons. The MLP was learned using only a single hidden layer and a small number of neurons, and it evaluates the variance ratio of the learnt BP utilizing holdout information.

[12] Another study investigates common practices in automobile threat modelling and ISO/SAE 21,434, as well as sensors and

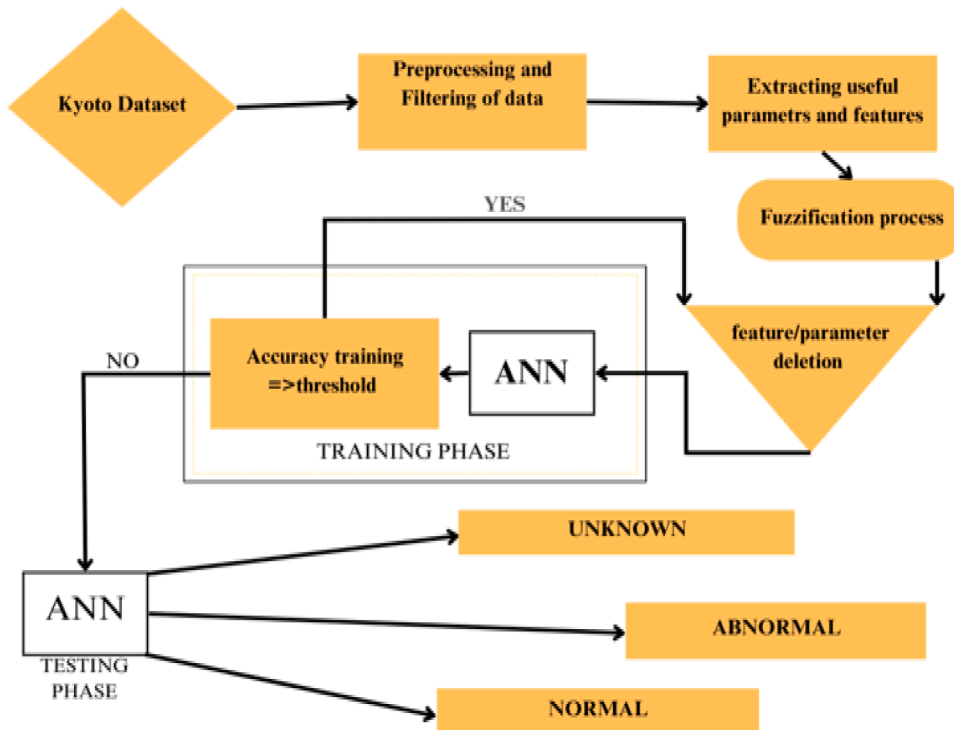


Fig. 2. Working of ANN with Kyoto Dataset.

Machine Learning (ML) techniques for AV-aware models and potential real-virtual threats. This study compares the hazard to AV perceived systems using the System Theoretical Process Analysis for Security (STPA-Sec) technique with ISO/SAE 21,434. Based on the findings, this study proposes a powerful mathematical modelling-based threat evaluation and assessment of hazards method for identifying cyber-physical threats to receiving systems. AV protocols are responsible for AVs' complex interactions and behaviours in their working design area [13].

However, future technological advancements should enhance the system's resilience and responsiveness in noisy environments. Additionally, integrating virtual reality operations with a recognizable self-directed coordinates approach could improve the system's capabilities. To achieve more accurate results, it is recommended to employ Intrusion Detection Systems (IDS) on Vehicular Ad-Hoc Networks (VANETs) with a virtual networking layer that exhibits superior disruption behaviour. This integration would leverage the benefits of IDS technology and the robustness of VANETs, leading to enhanced security measures in the Internet of Vehicles (IoV) domain.

4. Vehicle emissions

We can see from carbon emissions statistics that automotive emissions play a significant impact. The main reason to choose electric vehicles is to cut carbon emissions, although it is still far off. With the rise of AI, we may expect to see it used to identify, control, and propose the best solution to minimize car emissions.

[14] found that in the past, projections were produced using mathematical methods such as ARIMA, ANOVA, *t*-test and SARIMAX. To predict CO₂ emissions, the researcher employed randomized forest, decision tree model, and regression techniques. A special linear regression approach is used to predict sources with just one characteristic. Engine dimensions, fuel kind, piston count, car make, and model data are used for estimating CO₂ emissions. For assessing the impact of executed models, metrics for performance are applied. Using regression scores (R²-Score), MSE (Mean Squared Error), MAE (Mean Absolute Error), and other metrics, the reliability of each model is anticipated. The decision tree approach has a 95 % prediction accuracy compared to the random forest model's 99 %. According to further research and analysis, the random forest framework is the most effective AI model for predicting carbon dioxide emissions. [2] a Life Cycle Assessment (LCA) was used in the research based on nursery gas outflows may be the gold standard for determining the extent to which different options cut outflows throughout the vehicle's life and whether or not they are ecologically benign by nature. LCA might be a great tool, but it takes a lot of time, requires us to deal with many different factors, and can be very challenging. Even though this method is unconventional and necessitates testing and prior discussions, it shows a promising and straightforward solution to anticipate the correct nursery gas reserve funds for these elements immediately and sooner than the layout of the automobile parts.

A general AI that can analyze incomplete data, spot trends, and make predictions may be developed in the future. A capable prophetic AI will assist us in developing far more accurate natural judgements inside the complex division of emanations. Also, more complex and real data can be gathered to train, validate and test the model for robustness and accuracy.

5. Vehicle to vehicle communication

Society is moving towards autonomous automobiles, i.e., intelligent vehicles with data sharing. To create the future of automated traffic, every vehicle must be linked to the traffic network and other cars. Communication between cars is a new and increasingly artificial intelligence, machine learning demanding subject, and in the future, it will be a required step in vehicle development. But, to do this, we will need massive servers and a strong system in the cars to ensure that data interchange is safe and encrypted and there is no loss of information owing to changing environments, hacking, and overlapping radio waves. By involving intelligent radio, beam

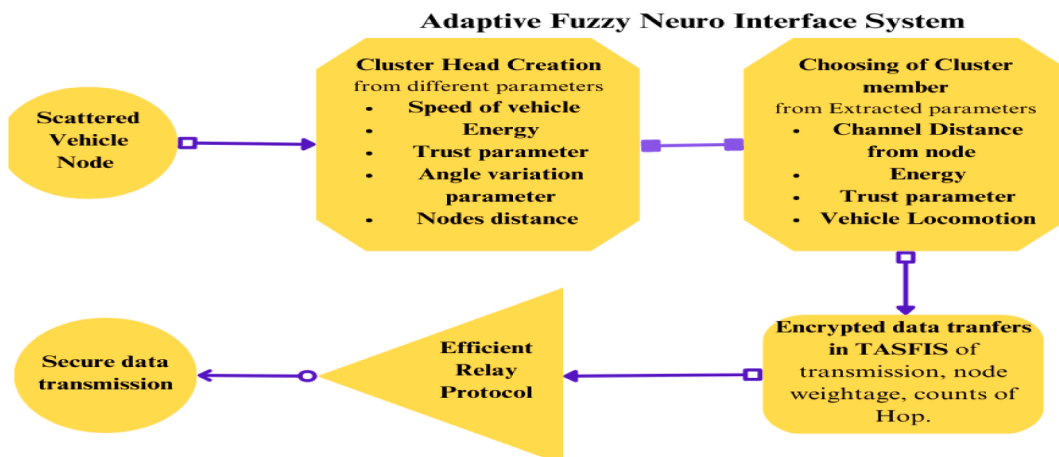


Fig. 3. Conceptual illustration of AI-TASFIS Flowchart.

formation, routing techniques, OFDM, and NOMA activities in an automobile network, the investigation expands on previous methods. However, the sole one was the algorithm and system developed for vehicle communication.

V2X can be enabled through one of two communication technologies. The first of these is called Dedicated Short-Range Communication (DSRC). The second uses the latest generation of (LTE) wireless communication and is known as telematics-V2X or C-V2X infrastructures. The ANN methods that can be used in V2X communication have all been compiled in this paper. These include Natural Language Processing, Game-Theoretic Learning, Genetic Algorithms, Inference, Fuzzy Logic, Automation, Swarm Intelligence, Expert Systems, Turning tests, Rational AI, Planning, Scheduling and Optimisation, and Machine Learning. As a consequence, this tactic results in accurate and transparent communications. The paper by [7] suggests using a Sugeno fuzzy inference framework based on machine learning (AI) to handle issues including security, private communication, and time-consuming delays. The trust-based security cluster solutions proposed are more successful at properly locating trustworthy nodes for communication, as well as having lower communication latency and overhead. Fig. 3 presents a conceptual illustration of the AI-TASFIS flowchart, depicting the sequential stages and processes involved in the AI-TASFIS framework.

The adjustable neural fuzzy deduction system, occasionally known as "ANFIS," is developed by integrating the Takagi Sugano fuzzy model with a neural learning network. These structures (ANNs) employ fuzzy logic and artificial neural networks. His-ANFIS, in conjunction with weighted training of neural network nodes, evaluates Sugano's fuzzy inference system involves precise instruction and tweaking of the language and numerical functions-based parameters of the member function. Fig. 4 depicts the ANFIS structure.

By solving the low-latency issue in V2V compared to V2I owing to the VD variability proportion, the RSC technique significantly improves system dependability in diverse IoV coordination. Through the localization of RSU using the RSU Migration model and optimization of the system's manufacturer acceptance rate, the DL-RSC solution attained an operational dependability rate of 71 %. Efficient RSU localization improved the consumer QoE by 67 %, while the multi-casting model utilizing linear code reduced the latency rate by 18 %. To improve the V2X communication, Open AI Reinforcement Learning Framework was employed, which served as the foundation for the simulation environment.

[15] use Coordinated Perception, which involves using Vehicle-to-Vehicle (V2V) interaction to propagate neural network-based traits across automobiles which shows great potential. However, the implications of Lossy Communication (LC) on feature sharing are not considered by existing cooperative perception algorithms. This research presents a unique approach that considers LC by setting up an LC-aware Restoration Network (LCRN) and a V2V Attention Mechanism (V2VAM) with interconnection-vehicular focus and ambiguity-aware inter-vehicular attention. The suggested method is tested on the public OPV2V dataset, and the recognition of three-dimensional objects using cloudy points. The outcomes presented that the suggested method outperforms cutting-edge cooperative perception algorithms and enhances detection performance in lossy V2V communication. Fig. 5 shows Synchronized perception with imperfect communication.

Comparing the result between ideal and lossy conditions with the implementation of the OPV2V dataset [16], in which the intermediate three fusion models trained with perfect communication like CoBEVT [17] and V2X-ViT [18]. This strategy has a good chance of successfully minimizing the harmful impacts of LC and enhancing relationships among ego vehicles alongside other vehicles. The framework derives useful visual feature maps. Each LIDAR feature extraction unit in a CAV is independent. The vehicle in question will get the nearby CAV feature maps following each characteristic extraction by CAV, and these acquired secondary characteristics will be incorporated into the remaining identification frameworks in the subsequent vehicle. The main part of our framework, the LC-Aware Repair Network, receives the secondary features gathered from other nearby CAVs and uses tensor-wise filtering to recover the intermediate feature map. The V2V Attention module then uses attention mechanisms to perform iterative inter- and intra-vehicle feature fusion. After getting the final fused feature map, box regression and classification are done, and the model generates the result.

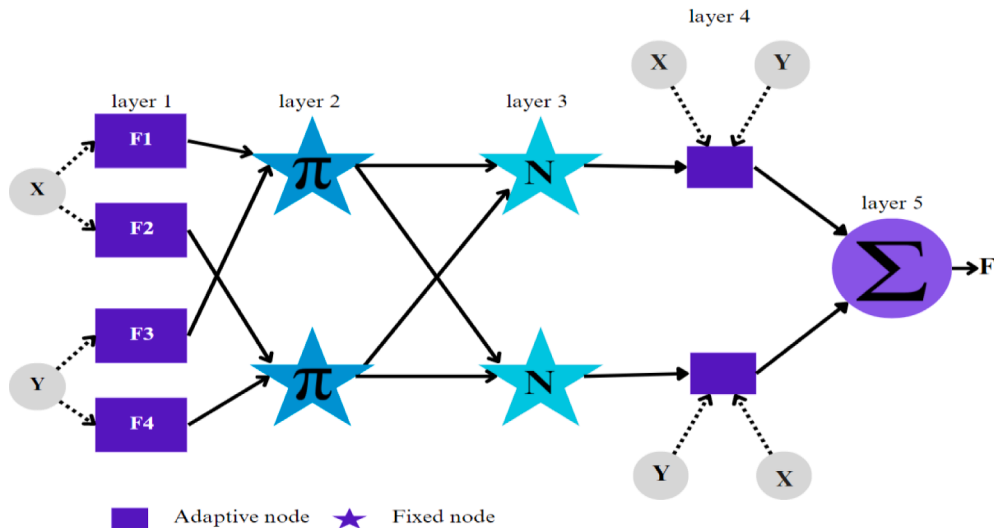


Fig. 4. ANFIS structure.

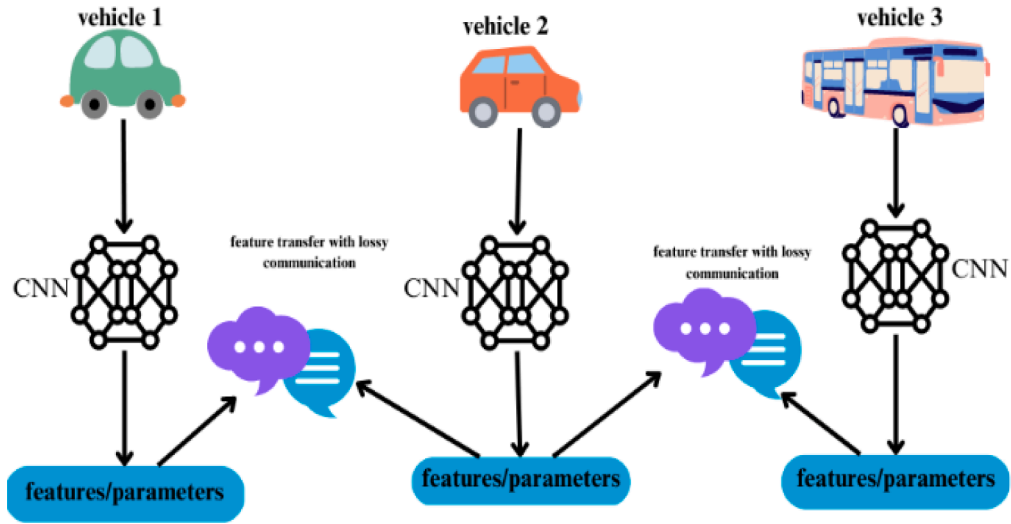


Fig. 5. Synchronized perception with imperfect communication.

Fig. 6 showcases an illustration of the LC-aware fusion framework.

The need to guarantee reliable data transmission is growing as networks become more dispersed and dynamic. [19] in this researchers attempted to use an algorithm, An interface-defined transportation network (SDN-FoG) built on fog computation, which aims to provide an adaptive vehicle network with efficient route prediction and low power consumption. The proposed architecture supports the scaleless topology since it is built on top of a hierarchy. The distributed grid solution is, therefore, as suggested. A complicated procedure is divided into multiple simpler ones. Each job is assigned to a class under the designated SDN-FoG, which laboriously completes the assignment. Three stages of this architecture are proposed: cloud computing level, SDN-FoG, and multi-media detection networking. [20] introduced a worldwide distributed computing system based on fog computing. Prioritized multicast queries were employed in scheduling and classification methods to ensure QoS requirements. Fig. 7 presents the conceptual architecture of SDN-FOG.

The security of the VANET network may be threatened by attackers who replicate or repurpose information. Serious problems like crashes or jams may result from this. [8] in this study, a confidence-establishing methodology called Enhanced Beacon Trust

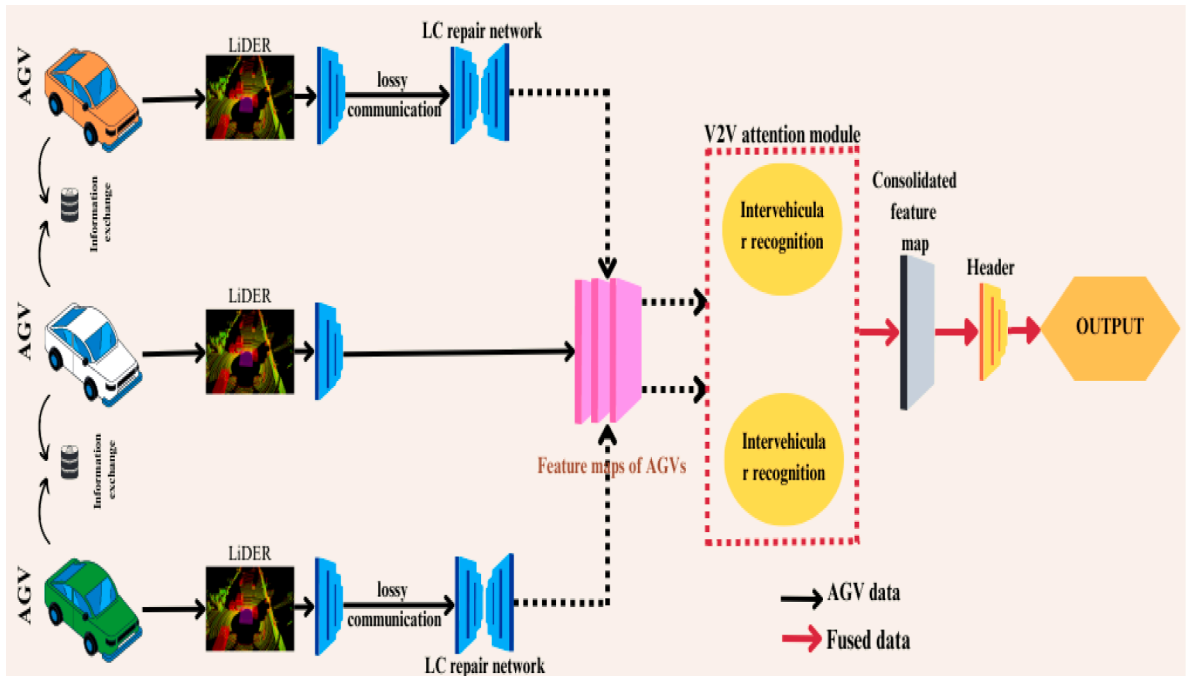


Fig. 6. Illustration of LC aware fusion framework.

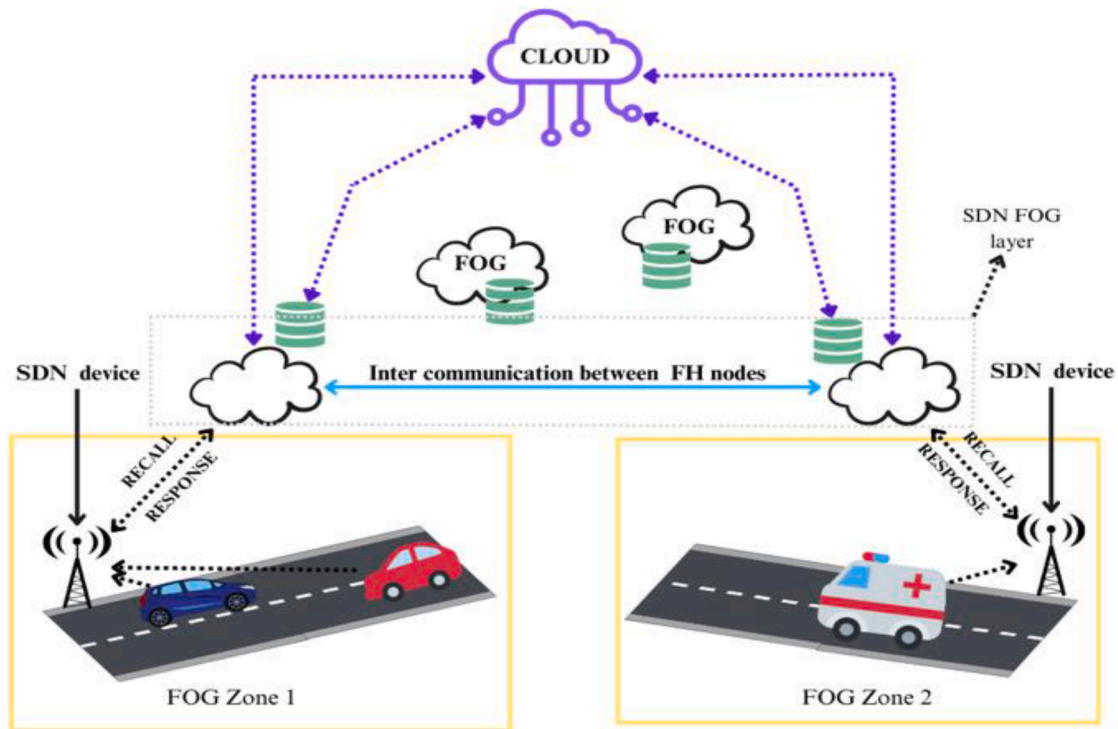


Fig. 7. SDN-FOG conceptual Architecture.

Management with Clustering Protocol (EBTM-CP) has been suggested to effectively identify malicious nodes and pick cluster chiefs. Fig. 8 depicts the architecture of an enhanced Beacon Trust Management system incorporating a Clustering Protocol. By fostering trust among the network's cars, this protocol's implementation intends to improve vehicle safety by enabling the transmission of dependable, accurate, and legitimate communications. Concerning package size, overall delay, packet corruption, PDR, utilization of energy, and performance, simulation findings demonstrate that EBTM-CP outperforms existing trust models, obtaining values of 297.99 Kbps, 46.34 %, 13 %, 165.6 Mbps, 67.49 %, and 16.34 %, respectively.

[21] study analyses the XAI models used in ICV surveillance systems (IDSs), their classifications, and available areas of study, demonstrating how XAI has the potential to significantly increase the effectiveness and acceptance of ICVs in the automotive sector. XAI was proposed to promote AI acceptance in some significant businesses and increase its transparency [22]. The main issue with connected items is their susceptibility to assaults and data manipulation during transmission over ICVs [23,24].

With the arrival of 6 G technology and stronger and more advanced AI, machine learning, CNN, etc., the connected AGV vehicle's future would be more robust and secure. Recent research paves the way for analyzing certain types of attacks using swarm intelligence techniques, analyzing how the suggested procedure is implemented on actual datasets, and predicting data transfer strategies using unsupervised machine learning techniques. Additionally, it is anticipated that fog computing models will reduce the communication time between applications and mobile devices.

Furthermore, considering multi-model Quality of Service (QoS) requirements from diverse fusion sources coupled with developing a protocol for analyzing network performance across multiple objectives, presents a significant challenge. However, these objectives can be achieved with thorough research and dedicated efforts. Despite the complexity, the proposed method holds great potential for revolutionizing service reliability in cooperative vehicular networks, ultimately leading to improved performance and heightened end-user satisfaction.

6. Driver monitoring system

The driver monitoring system employs artificial intelligence (AI) technologies to detect driver mental load and drowsiness to predict anomalies in driver behaviour before they occur, make decisions, and take necessary steps to control the situation ignorer to ensure the safety of that vehicle as well as the nearby vehicles. This AI implementation would make an excellent ADAS feature [25]. The whole tiredness identification process is expedited by the ocular motion-based technique's ability to recognize human faces and eyes more quickly than alternatives. Eye and facial recognition are 20 times faster than OpenCV when using data on a motion to trace the face, the mask comparison method, and the diamond searching technique.

In their study, [26] aimed to develop a cost-effective and real-time system capable of accurately detecting driver fatigue over an extended duration. To achieve this objective, the researchers utilized a camera to capture the facial frames of the driver. Subsequently,

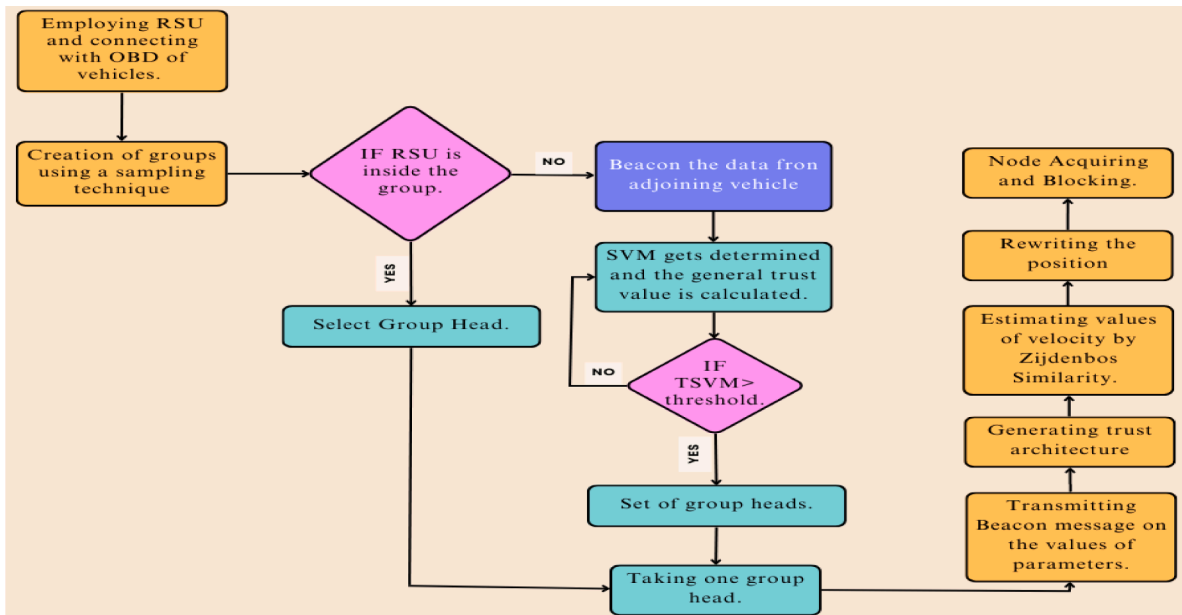


Fig. 8. Enhanced Beacon Trust Management with Clustering Protocol Architecture.

they employed image processing techniques to calculate both the eye-aspect ratio and mouth-opening ratio for each frame. By comparing these computed values to predetermined threshold values, the system could identify instances of driver fatigue. This innovative approach represents a variation in fatigue detection systems, offering a practical and efficient solution for monitoring driver tiredness over prolonged periods. Using the clip's extracted frames, the face is identified utilizing HOG with SVM algorithms. The system displays both positive and negative samples of test cases during the detection process. The recognized face is identified using the largest value result, and an outer boundary is drawn around it.

[27] study proposes a machine learning-based approach to lower the number of collisions among adolescents and novice drivers who could have ongoing medical issues that influence how they drive. The models were created using a LfO system called Falconet, which collected time-tagged records of the actions that 12 test participants in a driving simulation took. The models were then compared to the original traces to detect potentially dangerous medical conditions, such as Attention Deficit/Hyperactivity Disorder (ADHD). Extensive testing yielded favourable outcomes, with up to 82 % of test situations being accurately characterized. The method makes use of a programme called Falconet, which builds entities that replicate observable human vehicular behaviours using Neuro-evolution, Particle Swarm Optimisation and is dependent on context reasoning. Machine learning from observation is used to build models of human driving behaviour in individuals with ADHD. These models can be compared to the individual's actual driving behaviour to recognize when they are driving with uncontrolled ADHD conditions.

ADAS and IVIS systems can adjust their safety level based on changes in the driver's state and assess the time it takes the driver to react to deliver critical alerts at appropriate times.

7. Conclusions

This comprehensive review article explored AI applications beyond the automotive sector's Advanced Driver Assisting Systems (ADAS) framework. It highlighted the impact of AI on areas such as car emissions, predictive maintenance, security, connected vehicles, and driver monitoring systems. From algorithm like Game theory in vehicle security providing efficient and quick decision making to protect vehicle from the intrusion, fusing ANN and other methodologies like SVM, bee colony techniques to detect error. With reinforced learning in fusion with Machine Learning enhances the Predictive maintenance of the vehicle, using wavelet theory, Fourier transformation. XAI models, SDN-FoG computing provides quick encrypted connection between vhciles to safely tranfers data and communicate with vehicles. With the onset of the 6 g technology the latency in communication is nearer to zero which creates seamless V2V interaction. Blending A.I and computer image and video processing capabilities we can screen the driver mental and take safety decision whenever necessary.

The review identified gaps in the existing literature, emphasizing the need for further research to address these gaps and enhance future automotive systems. Integrating AI into automotive applications beyond ADAS opens up new possibilities for efficient and intelligent transportation systems. Additionally, the influence of mobility on network architecture, connectivity, and networking performance metrics was examined, providing insights into AGV technology, networking protocols, and swarms.

As the automotive industry continues to evolve, future advancements in AI and machine learning are expected to play a vital role in shaping the future of transportation. By addressing the identified gaps and building upon the existing research, experts can further enhance automotive systems' performance, efficiency, and safety, paving the way for a smarter and more connected future.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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